Another note on the simulation lemma

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Abstract

We prove the simulation lemma for discounted Markov decision processes (MDPs), which bounds how much the error in a learned transition model can affect the value of a policy. As a consequence, we see that planning in an accurate learned model yields a near-optimal policy.

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1 Motivation for model-based learning

Consider the (discounted) infinite-horizon MDP setting $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$. Assume that $\mathcal{S}, \mathcal{A}$ are finite, rewards are bounded in $[0, 1]$, the discount factor $\gamma \in [0, 1)$, and policies are stationary and stochastic. We have begun to see some online algorithms like TD learning methods, which learn value functions and policies in environments where the transition P and reward r are unknown. These algorithms are *model-free*, as they learn directly from experience without building a model of the environment.

With a *model-based* approach, we learn a transition model $\widehat{P}$ from collected data, then use planning algorithms like value iteration without further interaction with the environment. This can be desirable for many reasons. In chess, it is much easier to learn the rules and think ahead than to blindly learn from trial and error. In robotics, since data is expensive to collect, we would like to learn a model once and use it to plan for any number of policies and reward functions later.

Of course, our learned model $\widehat{P}$ will generally not be perfect. Small errors in predicting the next state can compound over time, leading to bad estimates of the value of a policy. We want some guarantee that if our simulated model $\widehat{P}$ is sufficiently close to the true model P, then the estimated value $\widehat{V}^\pi$ is close to the true value V^π. Moreover, we want the optimal policy $\widehat{\pi}^\star$ in our model to be near-optimal in the real world. This is the guarantee of the **simulation lemma** [1, 2]. To simplify our analysis, we assume throughout that we know the true reward r.

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2 State-action occupancy measures

For now, suppose we just have the true model P. As usual, given a policy π and an initial state s_0, we generate random trajectories $\tau = (s_0, a_0, s_1, a_1, \dots) \sim \rho^\pi_{s_0}$. Then, recall that the **step-h state-action occupancy measure** is defined by

$$\mathbb{P}^\pi_h(s, a|s_0) := \Pr_{\tau\sim\rho^\pi_{s_0}}(s_h = s, a_h = a).$$

In words, this is the probability that π reaches (s, a) from s_0 at time step h. This object is also commonly referred to as a **visitation distribution**, since it defines a distribution on $\mathcal{S} \times \mathcal{A}$.

It is also useful to take the *average* of the above probability over time, giving us a measure of how often π visits (s, a) throughout execution. In particular, we take a *discounted sum*, defining the **discounted state-action occupancy measure** by

$$d^\pi_{s_0}(s, a) := (1 - \gamma) \sum_{h=0}^{\infty} \gamma^h \, \mathbb{P}^\pi_h(s, a|s_0).$$

This is also a distribution on $\mathcal{S} \times \mathcal{A}$. Intuitively, by applying the discount γ^h, our sum gives more weight to time spent at (s, a) now rather than time spent at (s, a) far in the future. More mathematically, we require the discount factor so that our infinite sum converges, and we simply multiply by $(1 - \gamma)$ to ensure our distribution sums to 1. The definition of $d^\pi_{s_0}$ is motivated by the following attractive property.

Lemma 1. Let $f : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ be any function. Then,

$$\frac{1}{1-\gamma} \mathbb{E}_{(s,a)\sim d^\pi_{s_0}}[f(s, a)] = \mathbb{E}_{\tau\sim\rho^\pi_{s_0}}\left[\sum_{h=0}^{\infty} \gamma^h f(s_h, a_h)\right].$$

Exercise. Prove the above property.

3 The simulation lemma

We can now state the simulation lemma formally. Consider two infinite-horizon MDPs

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma), \qquad \widehat{\mathcal{M}} = (\mathcal{S}, \mathcal{A}, \widehat{P}, r, \gamma),$$

where we think of $\widehat{P}$ as an estimate of P. Note that r is identical. We will only write $d^\pi_{s_0}$ and $\rho^\pi_{s_0}$ in reference to the *true* model $\mathcal{M}$. Now, fix a policy π and initial state s_0, and let $V^\pi(s_0)$ and $\widehat{V}^\pi(s_0)$ be the exact value functions in the respective MDPs.

Theorem 2 (Simulation lemma)**.** The value error satisfies the identity

$$\widehat{V}^\pi(s_0) - V^\pi(s_0) = \frac{\gamma}{1-\gamma} \mathbb{E}_{(s,a)\sim d^\pi_{s_0}}\left[\mathbb{E}_{s'\sim\widehat{P}(\cdot|s,a)} \widehat{V}^\pi(s') - \mathbb{E}_{s'\sim P(\cdot|s,a)} \widehat{V}^\pi(s')\right]. \tag{1}$$

In words, the bracketed term in (1) measures the difference in the one-step (estimated) value after leaving (s, a), induced by the error in $\widehat{P}$. The simulation lemma relates the total error in $\widehat{V}$ to the average of this one-step prediction error over the state-action pairs that π tends to visit in the true MDP. In particular, to bound the value error, we only need $\widehat{P}$ to give accurate one-step predictions in regions covered by π. Loosening the RHS of (1), we also obtain a practical bound on the absolute error in terms of the L^1 error in $\widehat{P}$.

Corollary 3 (L^1 bound)**.** The value error is bounded absolutely by

$$\left|\widehat{V}^\pi(s_0) - V^\pi(s_0)\right| \le \frac{\gamma}{(1-\gamma)^2}\, \mathbb{E}_{(s,a)\sim d^\pi_{s_0}} \left\|\widehat{P}(\cdot|s,a) - P(\cdot|s,a)\right\|_1. \tag{2}$$

4 Proof of the simulation lemma

First, we will prove the identity in (1). In fact, we will prove the trajectory form of this identity, and then translate it via Lemma 1. Expanding $\widehat{V}^\pi(s_0)$ and $V^\pi(s_0)$ using the Bellman consistency equations and canceling the one-step reward,

$$\begin{aligned}
\widehat{V}^\pi(s_0) - V^\pi(s_0) &= \left(\mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\left[r(s_0,a_0) + \gamma\, \mathbb{E}_{s_1\sim\widehat{P}(\cdot|s_0,a_0)}\, \widehat{V}^\pi(s_1)\right]\right) \\
&\quad - \left(\mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\left[r(s_0,a_0) + \gamma\, \mathbb{E}_{s_1\sim P(\cdot|s_0,a_0)}\, V^\pi(s_1)\right]\right) \\
&= \gamma\, \mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\left[\mathbb{E}_{s_1\sim\widehat{P}(\cdot|s_0,a_0)}\, \widehat{V}^\pi(s_1) - \mathbb{E}_{s_1\sim P(\cdot|s_0,a_0)}\, V^\pi(s_1)\right].
\end{aligned}$$

Now, we add and subtract $\mathbb{E}_{s_1\sim P(\cdot|s_0,a_0)}\, \widehat{V}^\pi(s_1)$ (note the location of the hat) to obtain

$$\begin{aligned}
\widehat{V}^\pi(s_0) - V^\pi(s_0) = \gamma\, \mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\Big[&\mathbb{E}_{s_1\sim\widehat{P}(\cdot|s_0,a_0)}\, \widehat{V}^\pi(s_1) - \mathbb{E}_{s_1\sim P(\cdot|s_0,a_0)}\, \widehat{V}^\pi(s_1) \\
&+ \mathbb{E}_{s_1\sim P(\cdot|s_0,a_0)}\, \widehat{V}^\pi(s_1) - \mathbb{E}_{s_1\sim P(\cdot|s_0,a_0)}\, V^\pi(s_1)\Big],
\end{aligned}$$

which we rearrange to get

$$\begin{aligned}
\widehat{V}^\pi(s_0) - V^\pi(s_0) = \gamma\, \mathbb{E}_{a_0\sim\pi(\cdot|s_0)}&\left[\mathbb{E}_{s_1\sim\widehat{P}(\cdot|s_0,a_0)}\, \widehat{V}^\pi(s_1) - \mathbb{E}_{s_1\sim P(\cdot|s_0,a_0)}\, \widehat{V}^\pi(s_1)\right] \\
&+ \gamma\, \mathbb{E}_{a_0\sim\pi(\cdot|s_0),\, s_1\sim P(\cdot|s_0,a_0)}\left[\widehat{V}^\pi(s_1) - V^\pi(s_1)\right].
\end{aligned}$$

Note that the blue term looks very similar to the RHS of our desired identity, while the yellow term looks like the expression we started with, i.e., the LHS of our desired identity. In particular, we think of the first term in the above as the step $h = 0$ component in our trajectory, corresponding to the transition from s_0 to s_1. To derive the rest of the terms, we can recursively expand the yellow term with the same one-step expansion. Concretely, observe that

$$\begin{aligned}
\widehat{V}^\pi(s_1) - V^\pi(s_1) = \gamma\, \mathbb{E}_{a_1\sim\pi(\cdot|s_1)}&\left[\mathbb{E}_{s_2\sim\widehat{P}(\cdot|s_1,a_1)}\, \widehat{V}^\pi(s_2) - \mathbb{E}_{s_2\sim P(\cdot|s_1,a_1)}\, \widehat{V}^\pi(s_2)\right] \\
&+ \gamma\, \mathbb{E}_{a_1\sim\pi(\cdot|s_1),\, s_2\sim P(\cdot|s_1,a_1)}\left[\widehat{V}^\pi(s_2) - V^\pi(s_2)\right].
\end{aligned}$$

Plugging this back in,

$$\begin{aligned}
&\widehat{V}^\pi(s_0) - V^\pi(s_0) \\
&\quad = \gamma\, \mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\Big[\mathbb{E}_{s_1\sim\widehat{P}(\cdot|s_0,a_0)}\, \widehat{V}^\pi(s_1) - \mathbb{E}_{s_1\sim P(\cdot|s_0,a_0)}\, \widehat{V}^\pi(s_1)\Big] \\
&\qquad + \gamma^2\, \mathbb{E}_{a_0\sim\pi(\cdot|s_0),\, s_1\sim P(\cdot|s_0,a_0),\, a_1\sim\pi(\cdot|s_1)}\Big[\mathbb{E}_{s_2\sim\widehat{P}(\cdot|s_1,a_1)}\, \widehat{V}^\pi(s_2) - \mathbb{E}_{s_2\sim P(\cdot|s_1,a_1)}\, \widehat{V}^\pi(s_2)\Big] \\
&\qquad\quad + \gamma^2\, \mathbb{E}_{a_0\sim\pi(\cdot|s_0),\, s_1\sim P(\cdot|s_0,a_0),\, a_1\sim\pi(\cdot|s_1),\, s_2\sim P(\cdot|s_1,a_1)}\Big[\widehat{V}^\pi(s_2) - V^\pi(s_2)\Big].
\end{aligned}$$

Now, we have *two* terms that look like the RHS of (1), corresponding to steps $h = 0$ and $h = 1$, and another term that looks like the LHS of (1), but now as an expectation over s_2. Each time we expand, we obtain a new blue term, one step further along the trajectory and with one more factor of γ, and another leftover yellow term at the next state. Unrolling H times in total,

$$\begin{aligned}
&\widehat{V}^\pi(s_0) - V^\pi(s_0) \\
&\quad = \gamma \sum_{h=0}^{H-1} \mathbb{E}_{(a_0,s_1,a_1,\ldots,s_h,a_h)\sim\rho^\pi_{s_0}}\Big[\gamma^h\Big(\mathbb{E}_{s_{h+1}\sim\widehat{P}(\cdot|s_h,a_h)}\, \widehat{V}^\pi(s_{h+1}) - \mathbb{E}_{s_{h+1}\sim P(\cdot|s_h,a_h)}\, \widehat{V}^\pi(s_{h+1})\Big)\Big] \\
&\qquad + \gamma^H\, \mathbb{E}_{(a_0,s_1,a_1,\ldots,s_H)\sim\rho^\pi_{s_0}}\Big[\widehat{V}^\pi(s_H) - V^\pi(s_H)\Big].
\end{aligned}$$

To control the purple term, the inner expectations depend only on (s_h, a_h), so we can rename s_{h+1} to s' and take the outer expectations over entire trajectories $\tau \sim \rho^\pi_{s_0}$. Under a common expectation, we can move the sum inside via linearity. Taking the green term over trajectories as well,

$$\begin{aligned}
&\widehat{V}^\pi(s_0) - V^\pi(s_0) \\
&\quad = \gamma\, \mathbb{E}_{\tau\sim\rho^\pi_{s_0}}\Big[\sum_{h=0}^{H-1} \gamma^h\Big(\mathbb{E}_{s'\sim\widehat{P}(\cdot|s_h,a_h)}\, \widehat{V}^\pi(s') - \mathbb{E}_{s'\sim P(\cdot|s_h,a_h)}\, \widehat{V}^\pi(s')\Big)\Big] \\
&\qquad + \gamma^H\, \mathbb{E}_{\tau\sim\rho^\pi_{s_0}}\Big[\widehat{V}^\pi(s_H) - V^\pi(s_H)\Big].
\end{aligned}$$

Recall that value functions take values in $[0, 1/(1-\gamma)]$. As we take $H \to \infty$, the green term vanishes, since $\gamma^H \to 0$ and the expectation is bounded absolutely by $1/(1-\gamma)$. For the purple term, the h-th term of the sum is at most $\gamma^h/(1-\gamma)$ in absolute value, and so the series converges absolutely as $H \to \infty$. Thus, we obtain the trajectory form of the simulation lemma,

$$\widehat{V}^\pi(s_0) - V^\pi(s_0) = \gamma\, \mathbb{E}_{\tau\sim\rho^\pi_{s_0}}\Big[\sum_{h=0}^{\infty} \gamma^h\Big(\mathbb{E}_{s'\sim\widehat{P}(\cdot|s_h,a_h)}\, \widehat{V}^\pi(s') - \mathbb{E}_{s'\sim P(\cdot|s_h,a_h)}\, \widehat{V}^\pi(s')\Big)\Big].$$

Applying Lemma 1, we get the equivalent occupancy measure form in (1),

$$\widehat{V}^\pi(s_0) - V^\pi(s_0) = \frac{\gamma}{1-\gamma}\, \mathbb{E}_{(s,a)\sim d^\pi_{s_0}}\Big[\mathbb{E}_{s'\sim\widehat{P}(\cdot|s,a)}\, \widehat{V}^\pi(s') - \mathbb{E}_{s'\sim P(\cdot|s,a)}\, \widehat{V}^\pi(s')\Big].$$

Note that we could also have proved the occupancy form directly, by taking the outer expectations in the purple terms over just $(s_h, a_h) \sim \mathbb{P}^\pi_h(\cdot, \cdot|s_0)$. (How does this relate to the proof of Lemma 1?)

5 Proof of the L^1 bound

It remains to prove our inequality in (2), which translates bounds on $\widehat{P} - P$ to bounds on $\widehat{V} - V$.

Lemma 4. Let $p, q : S \to [0, 1]$ be discrete distributions, and let $f : S \to \mathbb{R}$ be bounded. Then,

$$\left|\mathbb{E}_{X\sim p} f(X) - \mathbb{E}_{X\sim q} f(X)\right| \le \sup_{x\in S}|f(x)| \cdot \|p - q\|_1.$$

Exercise. Prove the above property.

Now, we can prove our desired bound. Starting from the occupancy form in (1), taking the absolute value, and applying Jensen's inequality, we get

$$\begin{aligned}\left|\widehat{V}^\pi(s_0) - V^\pi(s_0)\right| &= \frac{\gamma}{1-\gamma}\left|\mathbb{E}_{(s,a)\sim d^\pi_{s_0}}\left[\mathbb{E}_{s'\sim\widehat{P}(\cdot|s,a)}\widehat{V}^\pi(s') - \mathbb{E}_{s'\sim P(\cdot|s,a)}\widehat{V}^\pi(s')\right]\right| \\ &\le \frac{\gamma}{1-\gamma}\,\mathbb{E}_{(s,a)\sim d^\pi_{s_0}}\left|\mathbb{E}_{s'\sim\widehat{P}(\cdot|s,a)}\widehat{V}^\pi(s') - \mathbb{E}_{s'\sim P(\cdot|s,a)}\widehat{V}^\pi(s')\right|.\end{aligned}$$

Note that $\widehat{V}^\pi$ is a function of $\mathcal{S}$, and it is bounded by $1/(1-\gamma)$. Applying Lemma 4 for the distributions $\widehat{P}(\cdot|s,a)$ and $P(\cdot|s,a)$, we obtain the bound in (2),

$$\begin{aligned}\left|\widehat{V}^\pi(s_0) - V^\pi(s_0)\right| &\le \frac{\gamma}{1-\gamma}\,\mathbb{E}_{(s,a)\sim d^\pi_{s_0}}\left[\sup_{s'\in\mathcal{S}}\left|\widehat{V}^\pi(s')\right| \cdot \left\|\widehat{P}(\cdot|s,a) - P(\cdot|s,a)\right\|_1\right] \\ &\le \frac{\gamma}{(1-\gamma)^2}\,\mathbb{E}_{(s,a)\sim d^\pi_{s_0}}\left\|\widehat{P}(\cdot|s,a) - P(\cdot|s,a)\right\|_1.\end{aligned}$$

6 Planning in the learned model

Finally, we return to our original motivation. Namely, we show that if $\widehat{P}$ is accurate, then planning in $\widehat{\mathcal{M}}$ gives us a good policy in $\mathcal{M}$. Let $\pi^\star$ and $\widehat{\pi}^\star$ be optimal in the respective MDPs.

Theorem 5 (Near-optimality of $\widehat{\pi}^\star$)**.** Suppose $\|\widehat{P}(\cdot|s,a) - P(\cdot|s,a)\|_1 \le \epsilon$ for all (s,a). Then,

$$V^{\pi^\star}(s_0) - V^{\widehat{\pi}^\star}(s_0) \le \frac{2\gamma\epsilon}{(1-\gamma)^2}, \quad \text{for all } s_0.$$

Proof. By adding and subtracting identical values, we can decompose the LHS as

$$V^{\pi^\star}(s_0) - V^{\widehat{\pi}^\star}(s_0) = \left[V^{\pi^\star}(s_0) - \widehat{V}^{\pi^\star}(s_0)\right] + \left[\widehat{V}^{\pi^\star}(s_0) - \widehat{V}^{\widehat{\pi}^\star}(s_0)\right] + \left[\widehat{V}^{\widehat{\pi}^\star}(s_0) - V^{\widehat{\pi}^\star}(s_0)\right].$$

By the assumption and (2), we know that $|\widehat{V}^\pi(s_0) - V^\pi(s_0)| \le \gamma\epsilon/(1-\gamma)^2$ for every policy π, which bounds the first and third terms. The second term is at most 0, since $\widehat{\pi}^\star$ is optimal in $\widehat{\mathcal{M}}$.

Here, we imposed the assumption that $\widehat{P}$ is accurate at *every* (s,a) uniformly, which is stronger than being accurate under any one policy's occupancy measure. By doing so, we use the simulation lemma to control the value errors for both $\pi^\star$ and $\widehat{\pi}^\star$, without assuming any knowledge of which state-action pairs they visit in advance.

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